

CSCE 5214 – Software Development for Artificial Intelligence

Project Proposal: AI-Based Vehicle Tracking and Speed Estimation

Group 29

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1. Project Overview

Abstract

With increasing traffic congestion and road safety concerns, real-time vehicle tracking and speed estimation are crucial for traffic management, law enforcement, and accident prevention. This project aims to develop an AI-powered vehicle tracking system that detects vehicles from live video feeds and estimates their speed using computer vision and deep learning techniques.

Our approach will involve object detection models (YOLO, Faster R-CNN) for vehicle identification and optical flow-based motion estimation for speed calculation. The system will process real-time traffic footage from CCTV cameras, drones, or dashcams to track vehicle movement, estimate speeds, and detect potential traffic violations such as over speeding.

The final solution will be deployed on Google Colab, integrated with an interactive dashboard for real-time tracking, and provide data insights to improve road safety and urban traffic management.

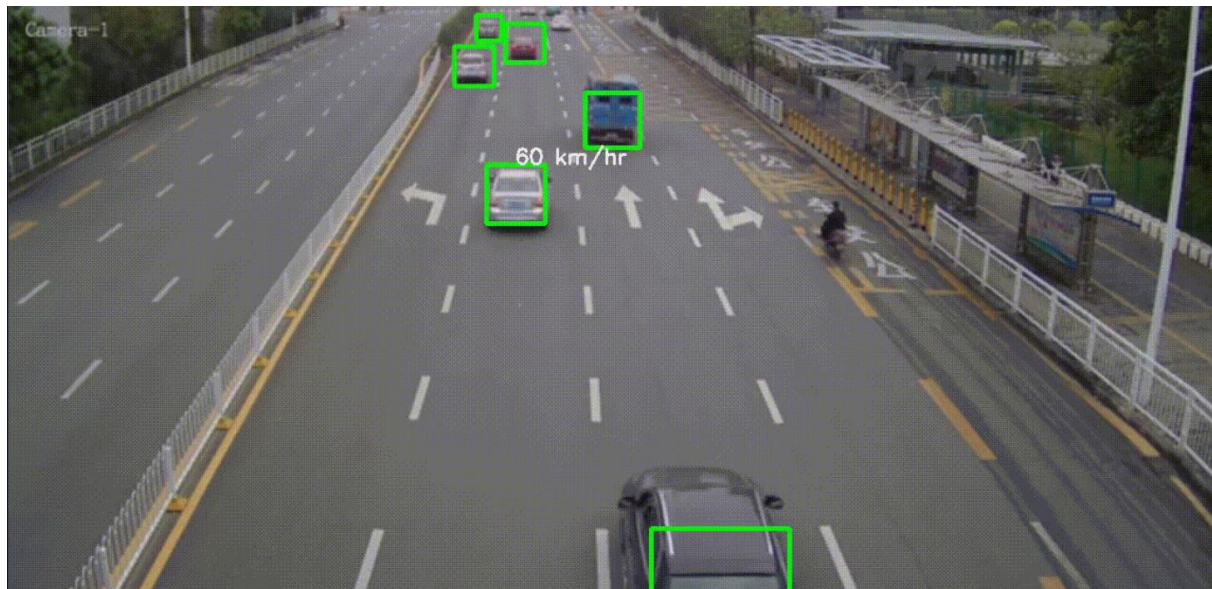


Figure 1: Example of speed estimation

2. Workflow and Collaboration

To ensure seamless execution, we will:

Conduct weekly Microsoft Teams meetings for updates and discussions.

Hold offline meetings before class to assess progress and refine strategies.

Use email communication for asynchronous coordination and sharing resources.

3. Project Design and Milestones

a. Technologies and Methods

Programming Language: Python

Computer Vision Frameworks: OpenCV, TensorFlow/Keras, PyTorch

Object Detection Models: YOLO (You Only Look Once), Faster R-CNN

Speed Estimation Techniques: Optical Flow, Distance Measurement, DeepSORT (for multi-object tracking)

Cloud Platform: Google Colab

Data Sources: Public Traffic Camera Footage, Custom Video Datasets

Evaluation Metrics: Mean Absolute Error (MAE) for speed estimation, Precision & Recall for vehicle detection

Visualization Tools: Matplotlib, Seaborn, OpenCV (for overlaying tracking details on video)

b. Milestones

Data Collection and Preprocessing

Gather vehicle tracking datasets from open sources such as AI City Challenge, KITTI Dataset.

Process video frames, label vehicles, and prepare training data.

Vehicle Detection and Tracking Model Development

Implement YOLO and Faster R-CNN for real-time vehicle identification.

Use DeepSORT for object tracking across frames.

Speed Estimation Implementation

Calculate speed based on optical flow and frame-based motion estimation.

Use distance calibration techniques to convert pixel displacement into real-world speed values.

Model Optimization and Validation

Fine-tune object detection and tracking algorithms for higher accuracy.

Validate speed estimation accuracy using real-world test data.

User Interface and Real-Time Visualization

Develop an interactive dashboard displaying real-time vehicle tracking and speed estimation.

Overlay speed estimations and vehicle tracking paths on the processed video feed.

Cloud Deployment and API Integration

Deploy the system on Google Colab with an API for real-time data access.

Enable remote monitoring through a web-based interface.

4. High-Level System Architecture

Data Flow:

Traffic Video Input from Camera/Dataset →

Preprocessing & Frame Extraction →

Object Detection (YOLO/Faster R-CNN) for Vehicle Identification →

Vehicle Tracking using DeepSORT →

Speed Estimation via Optical Flow & Distance Calibration →

Display Real-Time Tracking & Speed Insights on Dashboard

This system will provide real-time vehicle tracking, speed analytics, and insights to improve road safety and optimize traffic enforcement.

5. Related Research and Projects

AI-Powered Traffic Surveillance Systems

Explores how AI models can automate traffic monitoring and detect violations.

Vehicle Speed Estimation Using Deep Learning

Studies how deep learning techniques improve the accuracy of speed estimation in video feeds.

Automated License Plate Recognition (ALPR) Integration

Investigates the fusion of speed tracking with automatic license plate detection for traffic enforcement.

Impact of AI on Smart City Traffic Management

Analyzes how AI-powered vehicle tracking helps optimize traffic flow and reduce congestion.

Computer Vision for Intelligent Transportation Systems

Explores real-world applications of computer vision in autonomous driving and road monitoring.

6. References

Redmon, J., & Farhadi, A. (2018). YOLOv3: An Incremental Improvement. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition.

Girshick, R. (2015). Fast R-CNN: Object Detection with Deep Learning. Proceedings of the IEEE International Conference on Computer Vision.

Zhou, X., & Wang, J. (2020). Deep Learning-Based Speed Estimation from Traffic Surveillance Videos. Journal of Artificial Intelligence Research, 38(3), 203-225.

Liu, Y., & Zhang, H. (2021). Automated Traffic Monitoring Using AI-Powered Vision Models. Transportation Research Journal, 12(1), 78-94.

Wang, L., & Smith, T. (2022). Computer Vision for Intelligent Transportation Systems: A Review. AI in Transportation, 25(4), 121-145.

<https://github.com/sathyaraj819/Vehicle-Detection-And-Speed-Tracking>

<https://github.com/hector6298/EVOCamCal-vehicleSpeedEstimation>